Marketing Spend and Product Sales Analysis

1. Introduction

Marketing spend plays an important role in helping businesses promote their products and increase sales. Companies invest in multiple marketing channels, but understanding how these investments relate to product sales is equally important.

This project analyzes the relationship between marketing spend and product sales using a marketing dataset obtained from Kaggle. The analysis focuses on identifying patterns in the data and understanding how different marketing activities contribute to product sales.

2. Project Background

While dashboards can show marketing performance metrics, they do not always explain which marketing activities have the strongest influence on product sales.

To explore this further, regression analysis was used to study the relationship between marketing spend and product sales. The analysis was performed at both the channel level and category level to better understand how different marketing activities contribute to sales performance.

The goal of this project is to answer business questions using data and generate insights that can support marketing performance evaluation.

3. Business Questions

The analysis was designed around the following questions:

1) Does advertising spend influence product sales?
2) Would a more complex relationship explain product sales better?
3) Which marketing channels have the greatest impact on product sales?
4) Which marketing categories contribute most to product sales?

4. Dataset Overview

The dataset used in this project was obtained from Kaggle and contains information related to marketing spend and product sales.

The main variables included:

- TV Spend
- Billboards Spend
- Google Ads Spend
- Social Media Spend
- Affiliate Marketing Spend
- Influencer Marketing Spend
- Product Sold

To support broader analysis, marketing channels were grouped into higher-level categories.

5. Category Creation and Feature Engineering

To make the analysis easier to interpret, individual marketing channels were grouped into four categories.

Category	Channels Included
Traditional Category	TV, Billboards
Digital Category	Google Ads, Social Media
Affiliate Category	Affiliate Marketing
Partnership Category	Influencer Marketing

In addition, Total Spend was calculated by combining all marketing spend variables. This allowed the overall relationship between marketing investment and product sales to be analyzed.

6. Methodology

Four regression models were developed during the analysis.

Linear Regression

- Used to evaluate the overall relationship between total advertising spend and product sales.

Polynomial Regression

- Used to check whether a more complex relationship existed between spend and sales.

Multiple Regression (Channel Level)

- Used to identify the relative impact of individual marketing channels on product sales.

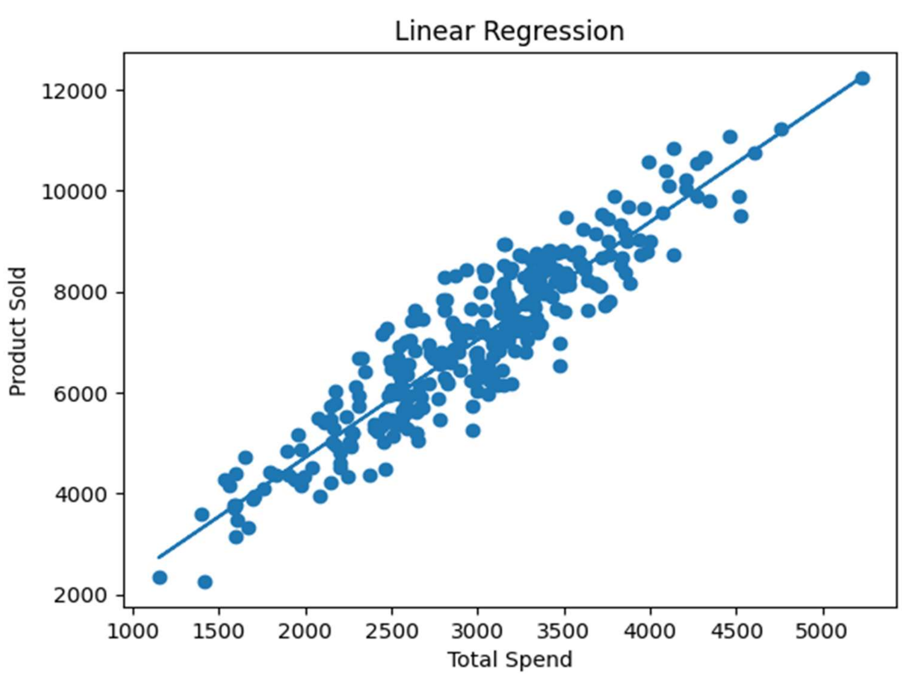
Multiple Regression (Category Level)

- Used to evaluate the contribution of broader marketing categories to product sales.

The performance of each model was evaluated using the R^2 metric.

7. Key Insights

a) Does advertising spend influence product sales?



Key Finding:

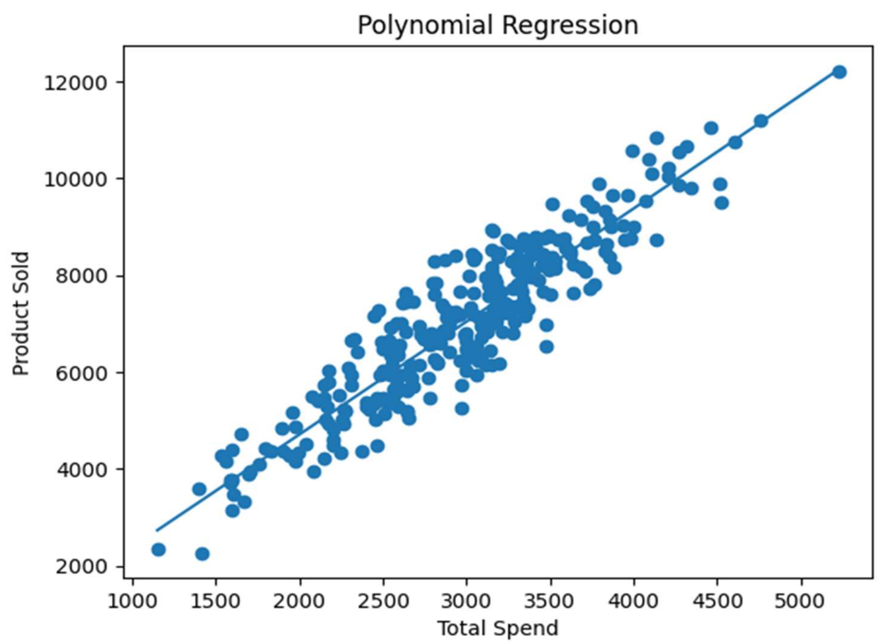
The Linear Regression model achieved an R^2 score of **0.847**, showing a strong relationship between advertising spend and product sales. This means that around **85% of the changes in product sales can be explained by total advertising spend** in the dataset.

Why It Matters?

This result shows that advertising spend plays an important role in influencing product sales. As spending increases, product sales also tend to increase, making advertising spend a useful factor to consider when evaluating sales performance and marketing effectiveness.

Note: An R^2 value closer to 1 indicates that the model is able to explain more of the variation in the target variable.

b) Would a more complex relationship explain product sales better?



Key Finding:

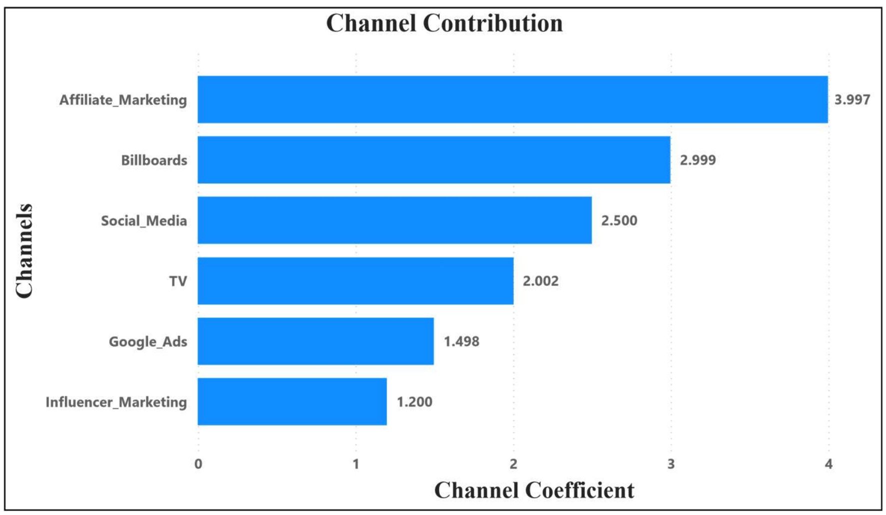
The Polynomial Regression model achieved an R^2 score of **0.847**, which was nearly identical to the Linear Regression model. Despite using a more complex approach, the model did not provide a noticeable improvement in explaining product sales.

Why It Matters?

This result suggests that the relationship between advertising spend and product sales is already well explained using a simple linear trend. Adding additional complexity to the model did not improve performance, making the simpler Linear Regression model easier to interpret and communicate.

Note: Polynomial Regression was used to test whether a curved relationship existed between advertising spend and product sales. Similar R^2 values indicate that the linear relationship already captures most of the pattern in the data.

c) Which marketing channels have the greatest impact on product sales?



Key Finding:

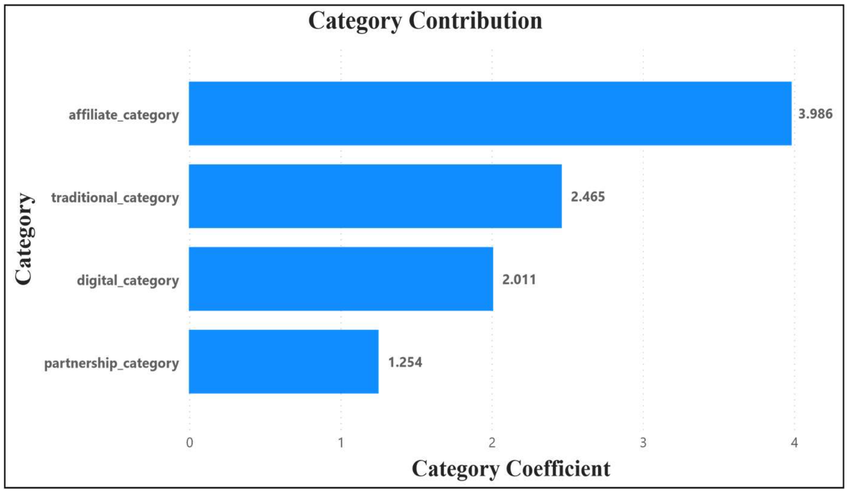
Among all marketing channels, Affiliate Marketing showed the highest impact on product sales, followed by Billboards and Social Media. Influencer Marketing showed the lowest impact in the model. This indicates that different channels contribute differently to overall sales performance.

Why It Matters?

When evaluating marketing activities, it is important to look beyond spending levels and understand which channels contribute more strongly to sales. This can help support future budget planning and channel evaluation.

Note: Channel coefficients represent the relative influence of each marketing channel on product sales. Higher values indicate a stronger contribution within the model.

d) Which marketing category contributes most to product sales?



Key Finding:

Affiliate Category showed the highest impact on product sales with a coefficient of 3.986, followed by Traditional Category (2.465) and Digital Category (2.011). Partnership Category showed the lowest impact among the categories analyzed. Similar to the channel-level analysis, Affiliate Marketing remained the strongest contributor to product sales, indicating a consistent pattern across both channel and category perspectives.

Why It Matters?

Analyzing marketing performance at the category level provides a broader view of how different groups of marketing activities contribute to sales outcomes. The results show that Affiliate Marketing demonstrates strong influence both as an individual channel and as a category, highlighting its importance within the overall marketing mix. This helps identify which areas are contributing more strongly to sales performance and can support future evaluation of marketing strategies.

Note: Higher coefficient values indicate a stronger estimated impact on product sales within the regression model.

8. Key Takeaways

The analysis identified a strong positive relationship between marketing spend and product sales.

- The analysis showed a strong relationship between marketing spend and product sales.
- Affiliate Marketing consistently showed the highest impact in both channel-level and category-level analysis.
- Linear Regression explained the relationship effectively, while Polynomial Regression did not provide additional value.
- Overall, the findings suggest that different marketing activities contribute differently to sales performance and should be evaluated individually.

9. Limitations

- The analysis was conducted using a publicly available Kaggle dataset.
- The results should be interpreted as insights from the dataset rather than direct business recommendations for real organizations.
- In addition, regression analysis helps identify relationships between variables but does not independently prove causation.

10. Learning Outcomes

Through this project, I gained practical experience in:

- Data preparation and feature engineering
- Regression modeling using Python
- Model evaluation using R^2 scores
- Business-focused data analysis
- Converting analytical findings into business insights
- Communicating results through reports and visual storytelling

This project also helped me understand the importance of connecting technical analysis with business questions rather than focusing only on model performance.

11. Conclusion

This project explored how marketing spend relates to product sales using multiple regression-based approaches.

By combining business understanding, statistical analysis, and insight generation, the project provided a clearer view of how different marketing activities contribute to sales performance.

The analysis demonstrated that data can be used not only to describe what happened but also to better understand the factors that may influence business outcomes.